

FPGA BASED REAL-TIME DETECTION PIPELINE FOR CHALLENGING ENVIRONMENTS

TABLE OF CONTENTS

1

1. Company Profile	2
2. Introduction	3
3. Problem Statement	4
4. Proposed Solution	5
5. Objective	9
6. Methodology	10
7. Progress	17
8. Video	18
9. References	19

COMPANY PROFILE

2

- **Established in 1964 by Nobel Laureate Abdus Salam**
- **Operates under a tripartite agreement with the Italian Government, IAEA, and UNESCO**



Science, Technology, and Innovation(STI) Unit:

Focus: Leverage science and technology for sustainable development

Activities:

- **Training programs and workshops in computer science, engineering, biotechnology, and renewable energy**
- **Promote innovative solutions and technological entrepreneurship**
- **Facilitate technology transfer**

Global Reach:

- **Provides access to high-level research facilities and collaborative opportunities**
- **Enhances scientific capacity in developing countries**

INTRODUCTION

3

Low visibility situations, caused by anything from heavy fog to a blinding sun, create a dangerous scenario for drivers.

The ability to judge distances, spot obstacles, and react to sudden changes becomes severely limited

A reliable system to detect objects hidden from plain sight becomes crucial for driver safety.



PROBLEM STATEMENT

4

Developing a robust and accurate object detection backbone on the edge for autonomous vehicles and other advanced driver-assistance systems (ADAS) operating in adverse weather conditions, particularly haze and fog.

CHALLENGES

- **FPGA implementation:** Memory Restrictions and static declaration introduce problems in writing programs.
- **Compression:** Compressing a model presents a daunting challenge. Shrinking its size while preserving performance demands careful optimization.

- **Limited Training Data:** Collecting and annotating a large dataset of images with diverse weather conditions can be expensive and time-consuming. Such a dataset is also not available.
- **Domain Shift:** Traditional object detection models trained on clear weather data struggle to generalize to hazy/foggy scenarios.

PROPOSED SOLUTION

5

Object detection backbone on an FPGA for faster Inferences

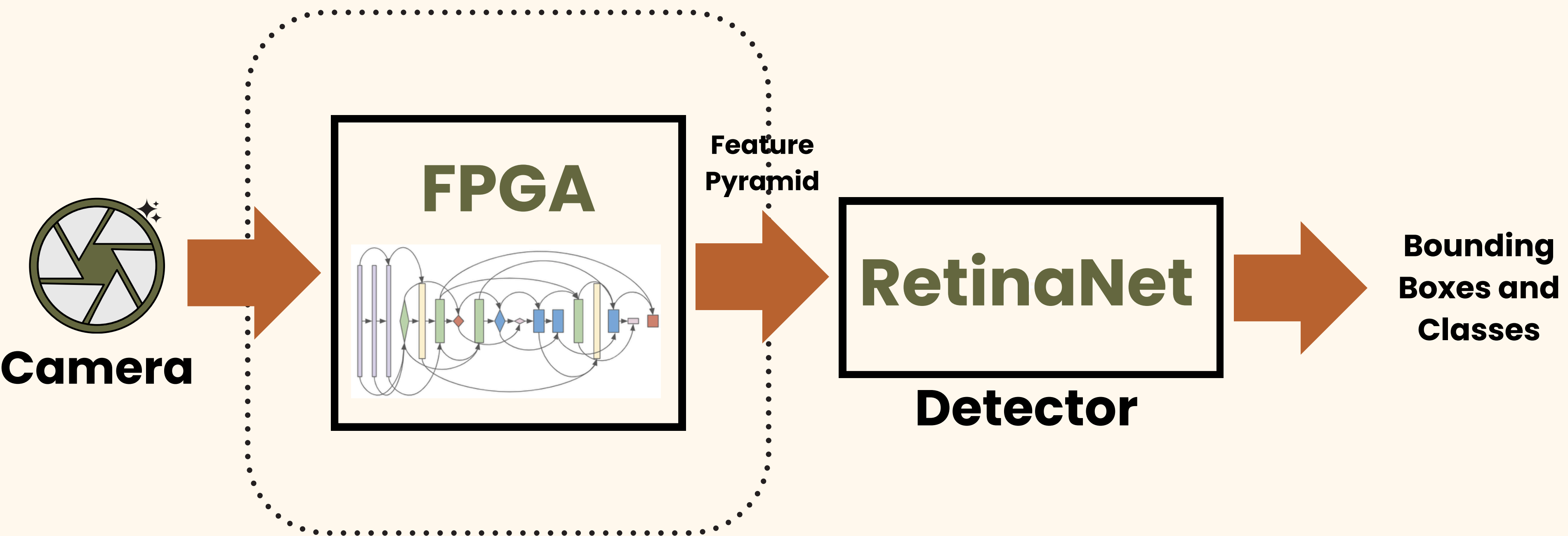


Figure 1. Complete Pipeline

PROPOSED SOLUTION

6

OBJECT DETECTION BACKBONE: SPINENET

Characteristic Features:

- Scale Permutations

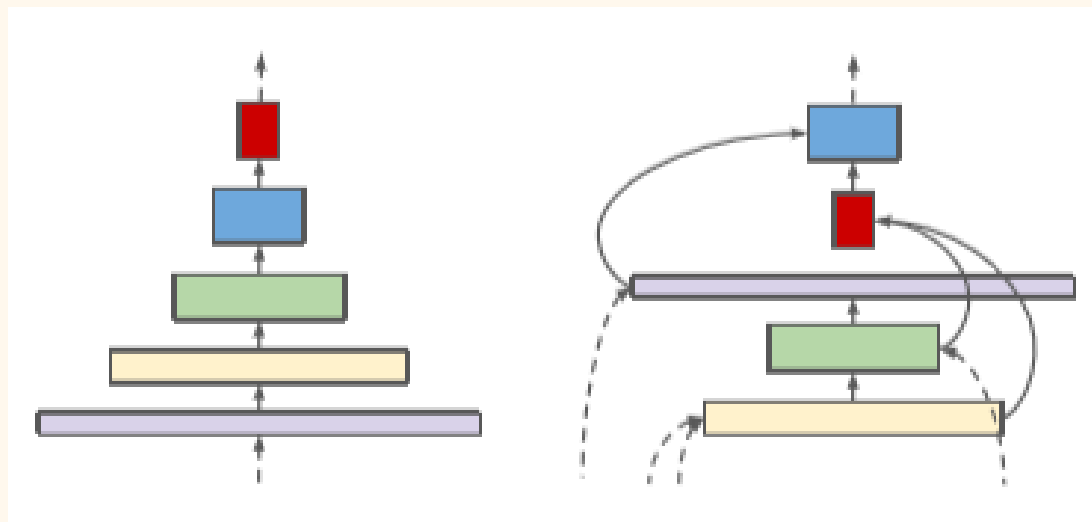


Figure 3. scale decreased vs permuted

- Cross Scale Connections
- Block Adjustments

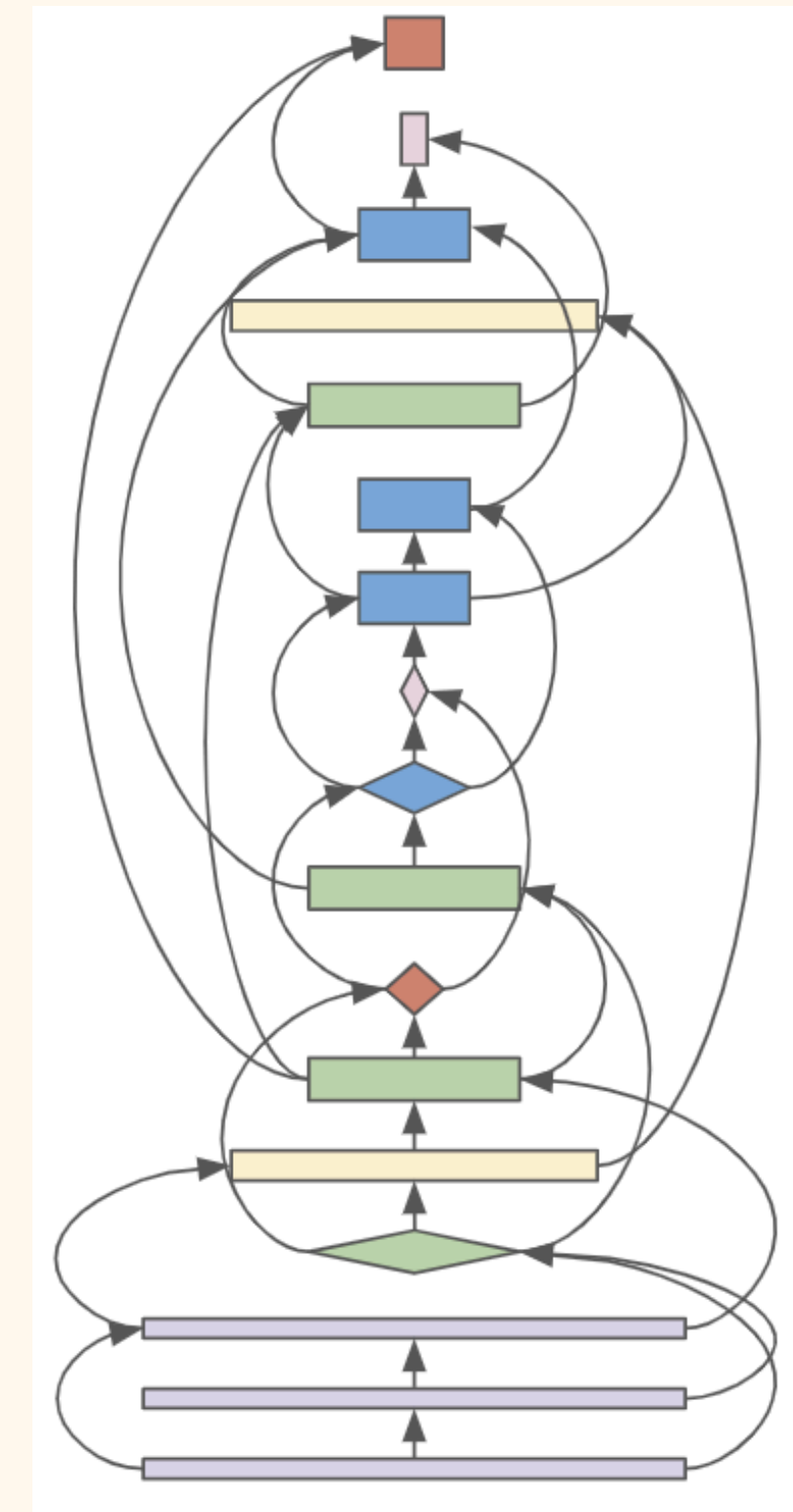


Figure 2. SpineNet 49/49S [1]

PROPOSED SOLUTION

7

WHY SPINENET?

- In spite of being trained on clear dataset, performs well on hazy dataset.
- Performs superior to Resnet50 on Clear data(Coco, Cityscapes)

- **Lesser FLOPS**

Resnet[2]	–	96.8B FLOPS
SpineNet49	–	85.4B FLOPS
SpineNet49S	–	33.8B FLOPS

- **Better Precision [1]**

Resnet	–	42.3 AP
SpineNet49	–	44.3 AP
SpineNet49S	–	41.5 AP

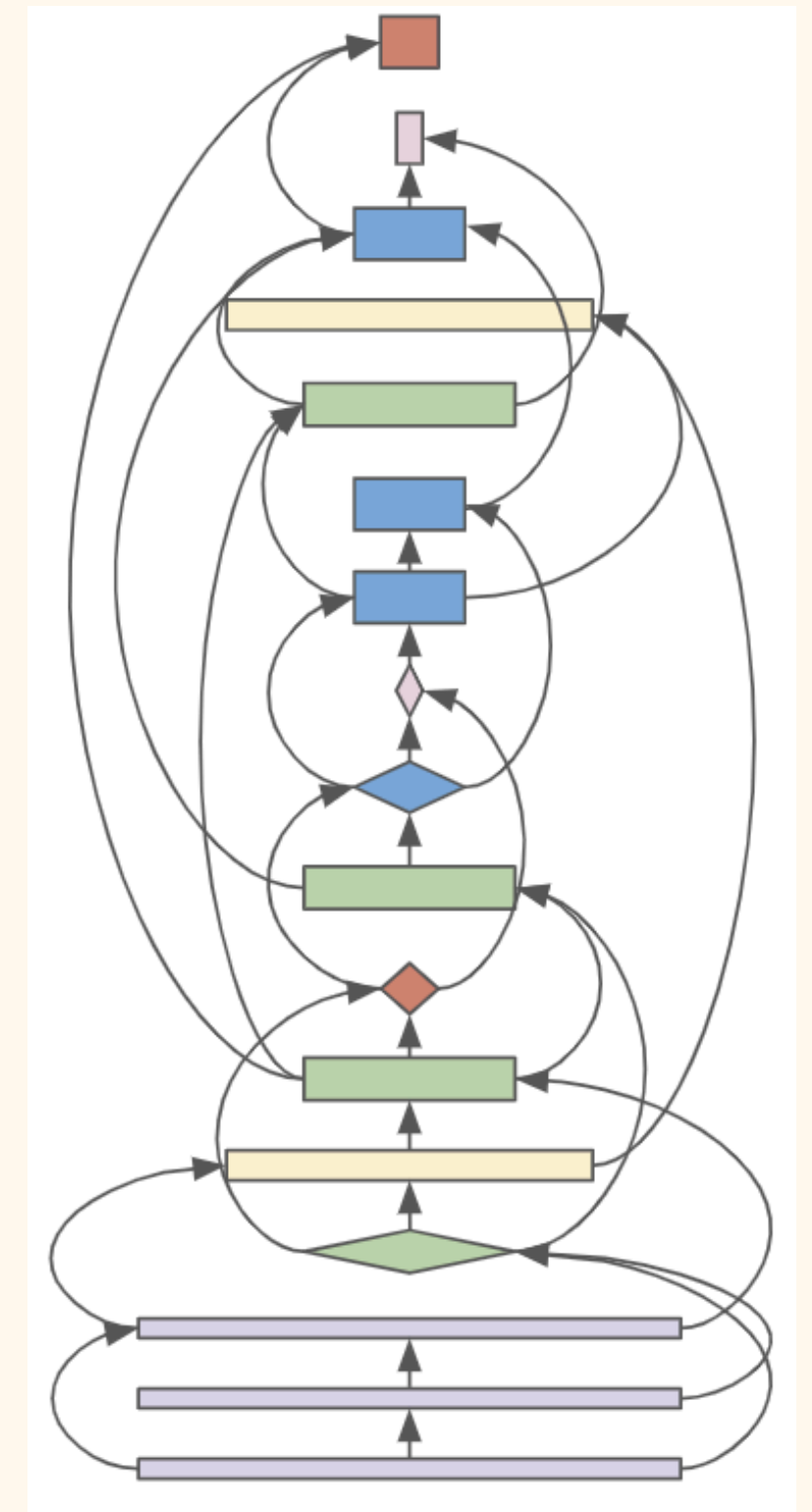


Figure 2. SpineNet 49/49S [1]

PROPOSED SOLUTION

8

DATASET:



- Cityscapes Dataset[2] for baseline MaP evaluation.
- Foggy variation of the dataset (leftImg8bit_trainval_foggyDBF) [3] for subsequent testing.



Synthetic haze of 3 intensities



Original

Only a subset of all classes,
Valid_Set = {car, bus,
motorcycle, person}, was
considered

Figure 3. Dataset

OBJECTIVES

9

1

To identify the appropriate SpineNet backbone architecture for deployment on FPGA:

By testing the various architectures on a synthetic hazy dataset[3] to create a performance baseline for comparison.

2

To export the SpineNet architecture to RTL using Vivado/Vitis HLS and optimize it for enhanced latency/throughput:

By implementing the SpineNet backbone on Vivado/Vitis and using various directives to optimize the code.

3

To experiment for possible compression of selected SpineNet architecture:

By using quantization, network pruning, knowledge distillation to reduce model size etc.[5].

OBJECTIVE 1:



To identify the appropriate SpineNet backbone architecture for deployment on FPGA:

- By
- ✓ Testing the architectures on Original Dataset
 - ✓ Testing the architectures on Hazy Dataset
 - ✓ Comparing metrics for the classes of interest (Valid_Set)

MODEL	MaP (%)
SpineNet 49	44.23
SpineNet 49S	40.65

Table 1: MaP for Valid_Set on original Data

MODEL	MaP (%)
SpineNet 49	36.91
SpineNet 49S	34.48
SpineNet 49MB	26.30

Table 2: MaP for Valid_Set on Foggy Data

MODEL	MaP (%)	Params (M)
SpineNet 49	31.53	28.5
SpineNet 49S	29.70	11.90
SpineNet 49MB	20.49	2.34

Table 3: MaP for all classes on Foggy Data

Outcome

Move forward with SpineNet 49S

Valid_set = {car, bus, motorcycle, person}

METHODOLOGY

11

OBJECTIVE 2:



To export the SpineNet architecture to RTL using Vivado/Vitis HLS and optimize it for enhanced latency/throughput:

APPROACH A: FROM SCRATCH USING VITIS HLS

By

- ✓ Creating a parser for TensorFlow checkpoint file.
- ✓ Using the parser to store the pre-trained weights as C++ header files.
- Implementing the SpineNet49S architecture using the header files and Vitis Vision Library / HLS Video Library.

APPROACH B: HLS4ML

By

- ✓ Designing the SpineNet49S architecture on Keras and exporting the h5 file.
- ✓ Using the hls4ml Package and the saved h5 file to generate HLS code

METHODOLOGY

12

Outcome :

LINK

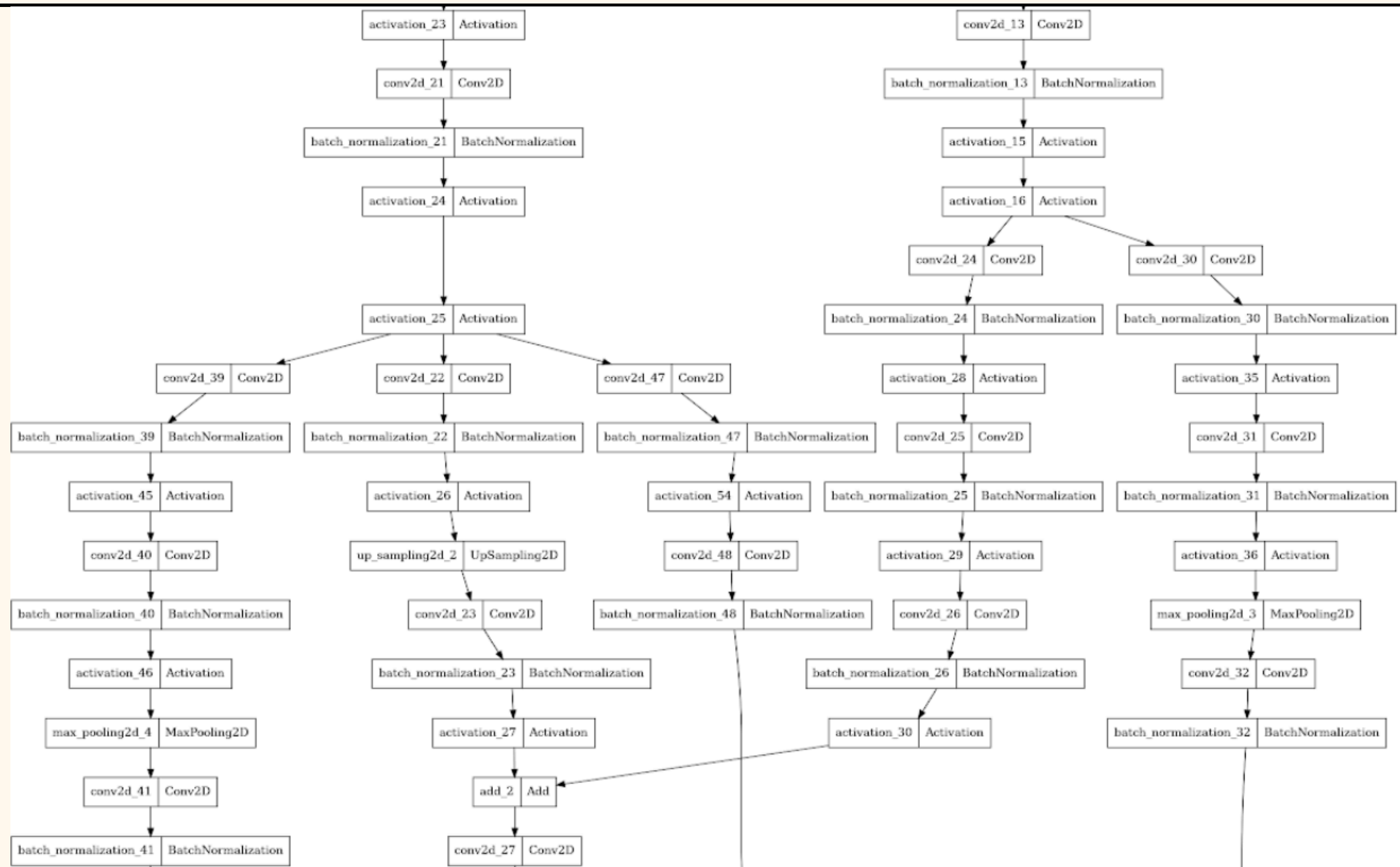


Figure 4.
SpineNet 49S

METHODOLOGY

13

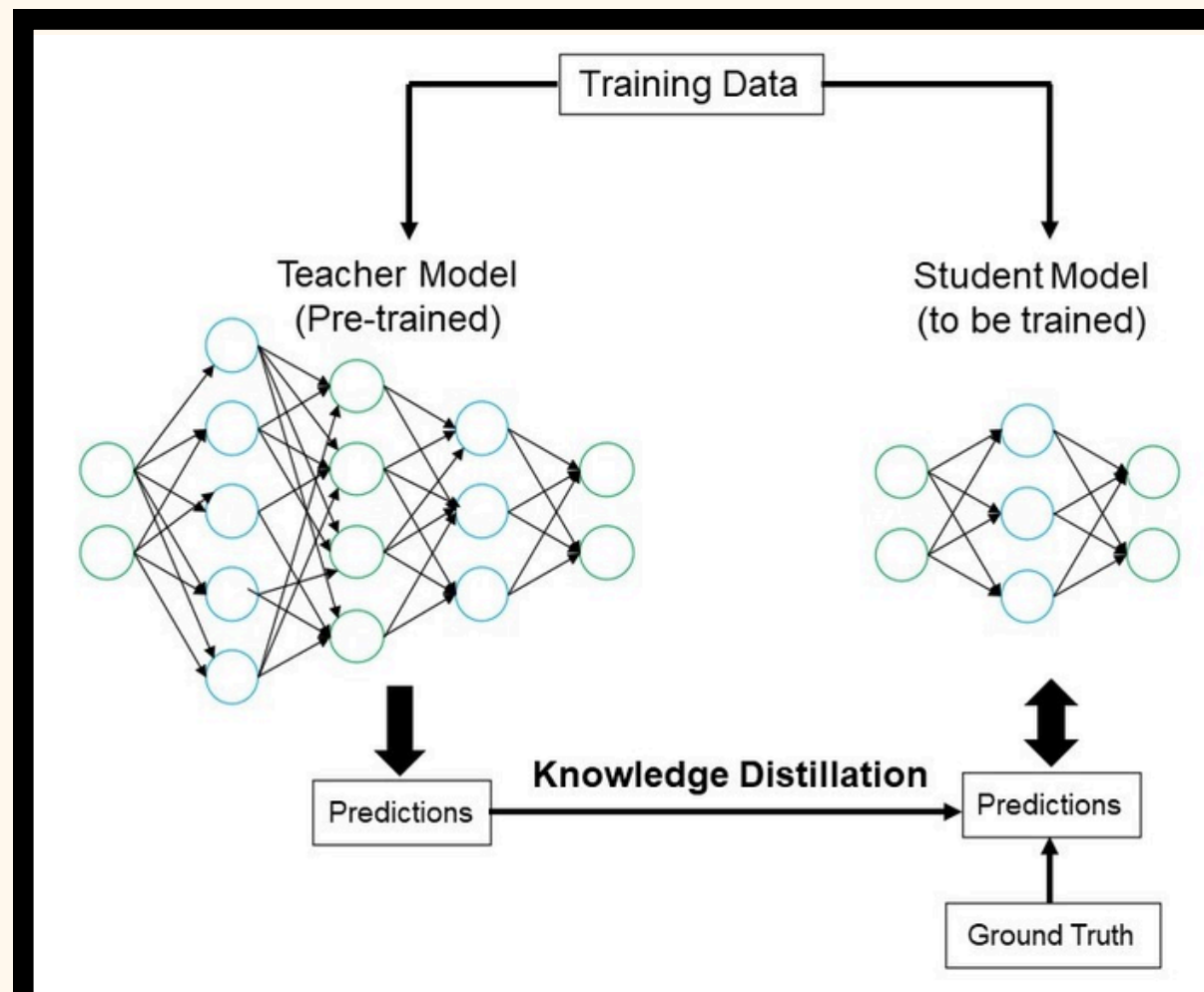
NAME	BRAM_18K	DSP48E	FF	LUT	URAM
DSP	-	-	-	-	-
Expression	-	-	0	86	-
FIFO	25149	-	360157	1325505	-
Instance	9	1	12068	19150	0
Memory	-	-	-	-	-
Multiplexer	-	-	-	1340	-
Register	-	-	206		-
Total	25158	1	372431	1346081	0
Available	624	1728	460800	230400	96
Utilization(%)	4031	0	80	584	0

Table 4. Utilization Chart

OBJECTIVE 3:



To experiment for possible compression of selected SpineNet architecture:



By

- 1.(Completed) Designing a student model
- 2.(Completed) Creating a Distiller with lssim loss function and SpineNet 49S as teacher.
- 3.(Completed) Executing the distiller to train the student on the Foggy Cityscape dataset.
- 4.(Completed) Evaluating the student's performance.
- 5.Repeating steps 1 to 5 if required.
- 6.(Completed) Re-implementing on HLS using hls4ml.

Figure 5.

KNOWLEDGE DISTILLATION

OBJECTIVE 3:



To experiment for possible compression of selected SpineNet architecture:

STRUCTURAL KNOWLEDGE DISTILLATION [6]

- **Distillation at Feature Pyramid Level**
- **Use of Statistical measures**
- **Patch-wise comparison**

$$l = \frac{2\mu_S\mu_T + C_1}{\mu_S^2 + \mu_T^2 + C_1}$$

$$c = \frac{2\sigma_S\sigma_T + C_2}{\sigma_S^2 + \sigma_T^2 + C_2}$$

$$s = \frac{\sigma_{ST} + C_3}{\sigma_S\sigma_T + C_3}$$

$$\ell_{SSIM} : \mathcal{L}_\epsilon = (1 - SSIM)/2 = (1 - (l^\alpha \cdot c^\beta \cdot s^\gamma))/2$$

OBJECTIVE 3:



To experiment for possible compression of selected SpineNet architecture:

Outcome :

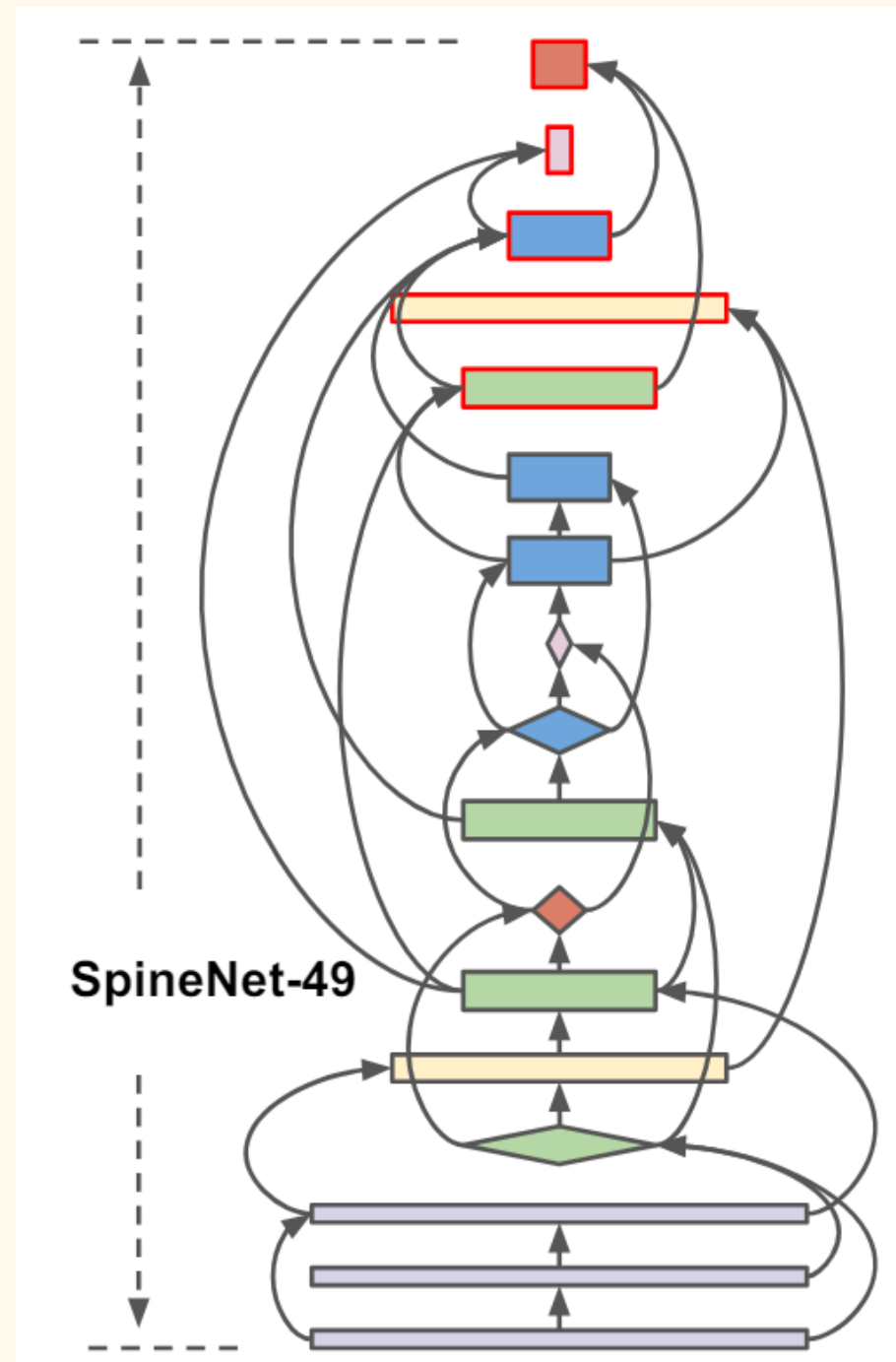


Figure 6. Teacher

Compression

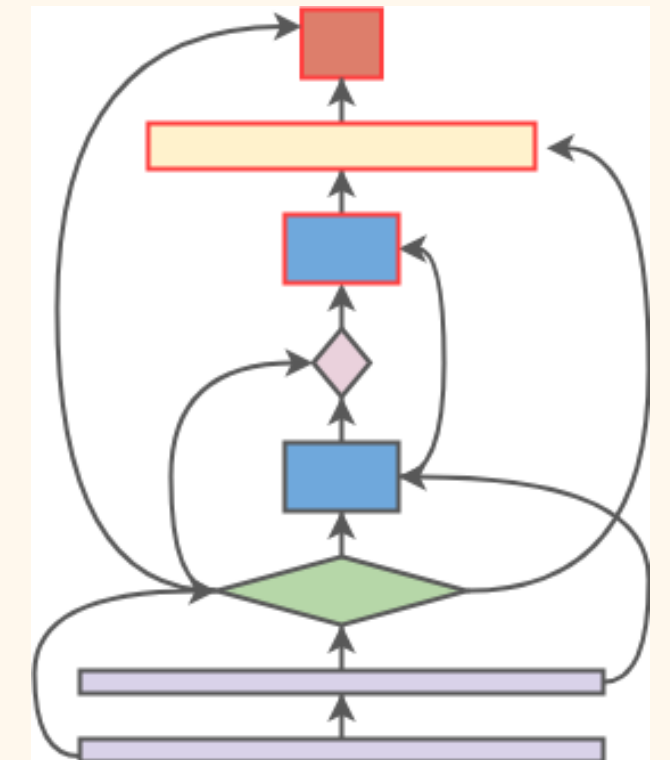
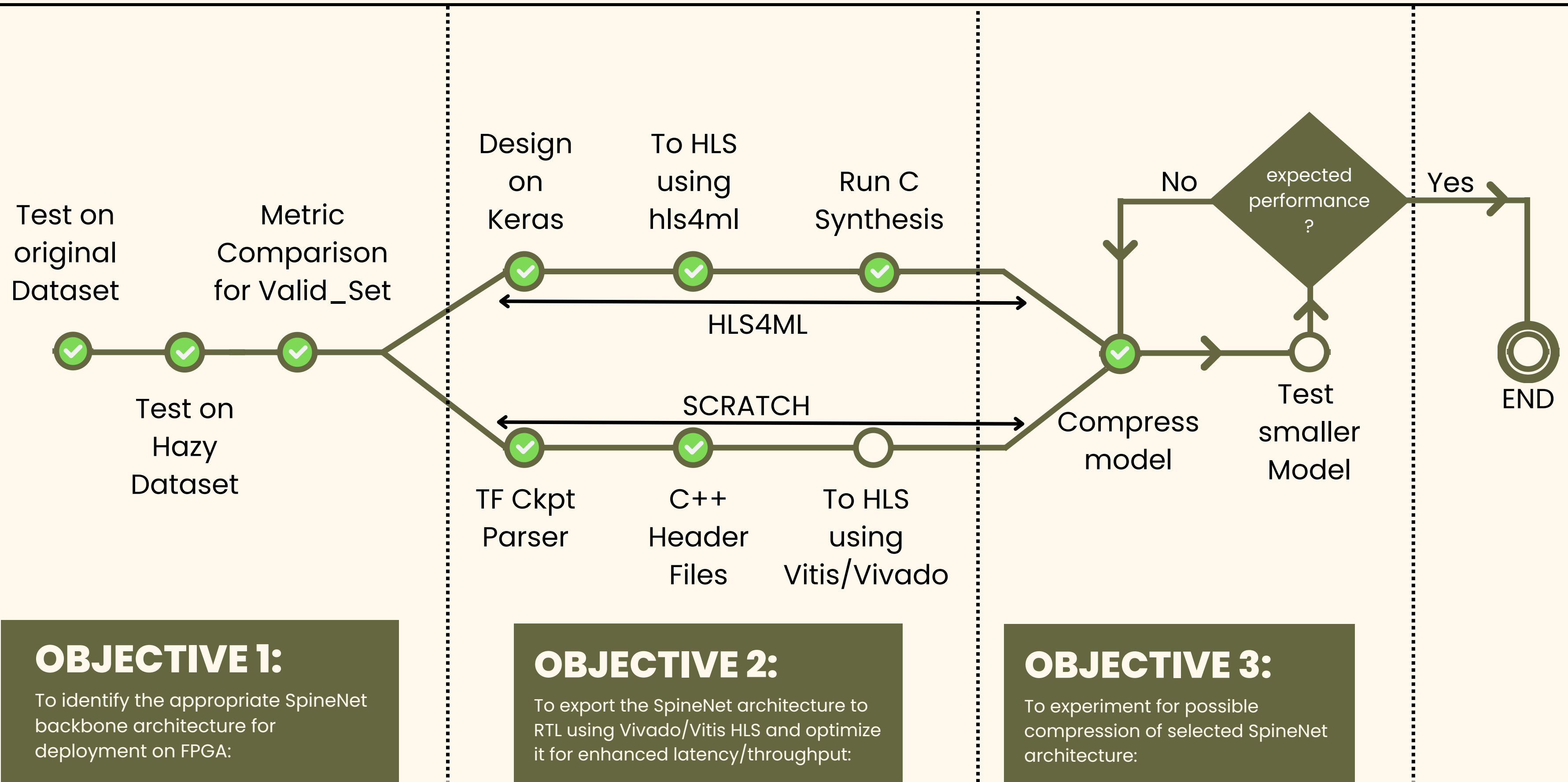


Figure 7. Student

PROGRESS

17



VIDEO LINK

18

<https://drive.google.com/drive/folders/1DQwwldugPAtjkouowxztyMC4nep3XeOD?usp=sharing>

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19

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ICTP, M-Lab



THANK YOU